Low-lying quasinormal modes of Kazakov–Solodukhin black holes from Chebyshev and continued-fraction methods

Adel H. Al-Yoorby[1,*]

[1]*Department of Physics, Khalifa University of Science and Technology, P.O. Box 127788, Abu Dhabi, United Arab Emirates*

ORCID: 0009-0000-1935-4392

Abstract. Black-hole quasinormal modes are damped resonances set by wave scattering from the curvature-generated potential barrier outside the horizon. We ask how the Kazakov–Solodukhin (KS) deformation changes these resonances at fixed mass. The minimally coupled scalar field is the cleanest physical probe because its propagation is fixed by the metric alone. Increasing the deformation moves the scalar barrier outward and makes its peak lower and less sharply curved. The lower barrier reduces the oscillation scale, while the flatter peak reduces the damping rate: the scalar ringdown therefore has a lower pitch and a slightly longer-lived envelope. For the quadrupolar $\ell = 2, n = 0$ mode at $a/M = 1$, $\mathrm{Re}\,\omega$, $-\mathrm{Im}\,\omega$, and $Q = \mathrm{Re}\,\omega/[2(-\mathrm{Im}\,\omega)]$ change by -7.29%, -2.59%, and -4.83%, respectively. Because the pitch falls more than the damping rate, the perturbation completes fewer cycles per damping time even though it decays slightly more slowly. We establish these shifts with Chebyshev collocation and a collocation-independent Frobenius continued fraction, accepting candidate roots only after residual, convergence, continuation, backward-error, and cross-method checks. In the Schwarzschild limit, the $N = 96$ scalar result and the continued-fraction value differ from the published benchmark by 5.60×10^{-11} and 8.24×10^{-13}. We report robust scalar fundamentals for $\ell = 2, 3, 4$, cross-validated first overtones at $N = 32$, and exploratory second overtones. The scalar pseudospectral basin broadens with deformation, indicating increased local numerical sensitivity without, by itself, implying an instability. For odd-parity gravitational perturbations we derive a gauge-invariant master equation under an inverse-Cowling closure; those axial frequencies remain conditional because the spherical KS action does not determine nonspherical quantum-source perturbations.

Keywords. Kazakov-Solodukhin black holes; quasinormal modes; Chebyshev spectral methods; Leaver continued fractions; spectral residual minimization; black-hole spectroscopy.

1 Introduction

After a black hole is disturbed, it does not vibrate indefinitely. Part of the disturbance falls through the horizon and part escapes to infinity, leaving a short-lived superposition of characteristic ringing modes. These quasinormal modes (QNMs) are resonances associated with an effective potential barrier generated by the spacetime geometry and angular structure. Their complex frequencies have an immediate physical meaning: $\mathrm{Re}\,\omega$ sets the pitch of the ringing, while $-\mathrm{Im}\,\omega$ sets the rate at which its amplitude decays. With the convention $e^{-i\omega t}$, writing $\omega = \omega_R - i\alpha$ gives a damping time $\tau = 1/\alpha$, where $\alpha > 0$. The spectrum therefore connects the geometry outside the horizon to black-hole stability, ringdown, and gravitational-wave tests of compact objects [8, 9, 10, 11].

*Corresponding author: alyoorby@gmail.com.

The first binary-black-hole detection made this regime observationally accessible [12]. Ringdown spectroscopy subsequently developed into a practical test of the Kerr hypothesis, with both angular harmonics and overtones entering the analysis [13, 10, 14]. Reliable spectra are therefore needed not only for Kerr black holes, but also for proposed quantum-corrected and exotic geometries whose ringdowns may depart from the general-relativistic expectation [11, 32].

QNM frequencies are obtained with several complementary tools. The Wentzel–Kramers–Brillouin (WKB) approximation treats the wave locally near the top of a slowly varying effective-potential barrier and connects the spectrum to its height and curvature. It is efficient when the multipole number exceeds the overtone number [21, 22, 23, 24, 11]. Direct time-domain evolution extracts the least-damped modes from waveforms, while Frobenius/Leaver continued fractions impose the frequency-domain boundary conditions through a minimal-solution recurrence [19, 20]. The asymptotic iteration method provides another recursive treatment of the radial ordinary differential equation [25]. Spectral and pseudospectral methods discretize the boundary-value problem globally and often return many eigenvalues in a single solve [28]. Each technique has a different failure mode: WKB expansions are least reliable for low multipoles and high overtones, waveform fits favor the least-damped tones, and finite spectral pencils may contain spurious roots [24, 11, 28]. We therefore use the Chebyshev spectrum as a starting point and judge each mode with convergence, continuation, residual, and continued-fraction tests.

The background considered here is the quantum-deformed Schwarzschild metric of Kazakov and Solodukhin, obtained from a two-dimensional dilaton-gravity description of spherically symmetric quantum fluctuations [15]. Its global interpretation requires some care—the deformation improves aspects of the central geometry without generically satisfying every criterion used for a regular black hole [16]—but it provides a well-defined static metric on which wave propagation can be studied. Earlier calculations found modest changes in the fundamental QNMs and a much stronger response in parts of the overtone spectrum [17, 18]. The central physical question is consequently simple: at fixed mass, does increasing a/M make the exterior potential barrier higher or lower, and how does that change the pitch and lifetime of the trapped waves? The Schwarzschild limit supplies the numerical control experiment; the finite-a/M branches answer the physical question.

QNMs are non-self-adjoint resonances, and this matters numerically. Outgoing boundary conditions lead to complex spectra that may be sensitive to both operator perturbations and discretization [32, 33]. We write the finite Chebyshev problem as $P_N(\omega)u = 0$ and monitor $\sigma_{\min}[P_N(\omega)]$. Equivalently, $\sigma_{\min}[P_N(\omega)] = \sqrt{\lambda_{\min}[P_N(\omega)^\dagger P_N(\omega)]}$; the eigenvalue itself is the square of the singular-value residual. Singular-value tests are standard tools for nonlinear eigenvalue problems [34], while related resolvent and pseudospectral diagnostics have become increasingly important in black-hole perturbation theory [32, 33]. Here they are used to decide whether a root of the generalized Chebyshev pencil is a converged KS resonance. The Frobenius calculation supplies a numerically distinct, collocation-independent discretization of the same radial equation and is particularly useful for following a mode as a/M changes [19, 18].

Modes are labelled by the spherical-harmonic multipole ℓ and the overtone index n. The least-damped mode for a fixed ℓ is the fundamental, $n = 0$; successively more strongly damped modes are the overtones $n = 1, 2, \ldots$. Spherical symmetry makes the frequency independent of the azimuthal label m, which is suppressed. Here "low-lying" means the fundamental and first few overtones nearest the real-frequency axis. Our scope is this low-lying KS spectrum.

The scalar $n = 0$ modes are robust fundamentals; scalar $n = 1$ branches are cross-validated low-lying first overtones at $N = 32$; $n = 2$ entries are exploratory branch diagnostics; and higher overtones are outside the scope. Recent analytic-continuation methods based on confluent-Heun solutions explicitly target complete and highly damped spectra of type-D black holes [36]; the present calculation is not intended to compete in that regime. The minimally coupled scalar field gives the cleanest physical probe of the KS metric because it propagates without perturbing the effective source that supports the geometry. The axial, or odd-parity, sector instead describes metric perturbations with parity $(-1)^{\ell+1}$. For that sector we derive the gauge-invariant equation and then impose a frozen-source, inverse-Cowling closure, as in perturbative treatments of general spherical backgrounds [2, 3, 4]. Those frequencies characterize the stated effective model, not a unique nonspherical completion of the KS quantum source.

1.1 Literature context and related work

1.1.1 Quasinormal modes and black-hole spectroscopy

The theory of black-hole perturbations begins with the Regge–Wheeler analysis of Schwarzschild perturbations [1] and early numerical recognition of damped black-hole ringing [8]. Modern reviews emphasize that QNM spectra provide information about stability, boundary conditions, late-time behavior, and possible observational tests of general relativity [9, 10, 11]. In gravitational-wave astronomy, QNM spectroscopy is especially important because detecting multiple ringdown tones can test whether the remnant is described by the Kerr family [13, 14].

1.1.2 Quantum-corrected Schwarzschild geometries

Kazakov and Solodukhin proposed a quantum deformation of the Schwarzschild solution in which the radial center is displaced to a Planck-scale surface through a spherically symmetric quantum correction [15]. Later geometric analyses have clarified that the original construction improves some regularity properties but does not automatically remove all curvature singular behavior in the sense usually meant in general relativity [16]. Phenomenological work on KS black holes has studied QNMs, scattering, shadows, Hawking radiation, and overtone sensitivity [17, 18]. These studies motivate higher-accuracy spectral calculations and systematic comparisons across deformation parameters.

1.1.3 Spectral approaches to QNM computation

Spectral methods are attractive for QNM problems because they approximate smooth functions globally and can converge rapidly when the boundary conditions and coordinate compactifications are chosen well. Leaver's Frobenius continued-fraction method remains a standard benchmark for black-hole QNM calculations [19]. Jansen's QNM spectral package demonstrates how differential QNM equations can be discretized and solved as generalized eigenvalue problems over a wide class of asymptotics [28]. More recent studies by Batic, Dutykh, and collaborators use spectral formulations to revisit noncommutative Schwarzschild black holes, Lee–Wick black holes, and Morris–Thorne wormholes, correcting or sharpening conclusions obtained from lower-order approximation methods [29, 30, 31]. Pseudospectral and pseudospectrum analyses also highlight that QNM spectra can be sensitive to operator perturbations, an important caution for finite-resolution spectral discretizations [32]. Cao et al. recently applied this pseudospectrum perspective to a quantum-corrected Schwarzschild black hole, which is especially relevant background for the present KS calculation [33].

1.1.4 Residual and singular-value viewpoints

For a nonlinear spectral operator $P(\omega)$, an exact QNM satisfies $P(\omega)u = 0$ for a nonzero state u. At finite resolution this condition can be tested through the smallest singular value, while $P^{\dagger}P$ expresses the same defect as a positive-semidefinite Hermitian problem. Smallest-singular-value methods are well established for nonlinear eigenvalue problems [34], and closely related diagnostics occur in pseudospectral studies of non-self-adjoint black-hole operators [32, 33]. We use this viewpoint to refine the KS roots and then compare them with spectral convergence and a Leaver continued fraction [19, 28].

2 Methods

2.1 Kazakov–Solodukhin perturbation problem

Kazakov and Solodukhin obtained a spherically symmetric quantum deformation of the Schwarzschild solution by reducing the four-dimensional Einstein theory to an effective two-dimensional dilaton-gravity problem and choosing a renormalizable dilaton potential [15]. We use the areal-radius form subsequently employed in the KS QNM analysis of Konoplya [17]:

$$\begin{aligned} \mathrm{d}s^2 &= -f_a(r)\mathrm{d}t^2 + f_a(r)^{-1}\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2, \\ f_a(r) &= \frac{\sqrt{r^2 - a^2}}{r} - \frac{2M}{r}, \end{aligned} \tag{1}$$

Here r is the areal radius, $M > 0$ is the mass parameter, and $a \geq 0$ is a length scale. The areal radius is defined so that a sphere at coordinate r has area $4\pi r^2$. The function f_a is the lapse: it controls the redshift of static clocks and vanishes at the event horizon. No radial-coordinate transformation relative to the convention of Ref. [17] is made. In that convention the present numerical work treats a directly as the deformation parameter; no otherwise unused renormalized coupling is introduced. Reality of the metric coefficients restricts the coordinate patch to $r \geq a$. The outer horizon follows from $f_a(r_h) = 0$:

$$\sqrt{r_h^2 - a^2} = 2M, \qquad r_h = \sqrt{4M^2 + a^2}. \tag{2}$$

Thus $r_h > a$ for every finite a/M, and the value $a/M = 1$ used below is not an extremal or degenerate-horizon limit; it is the upper endpoint of the deformation interval sampled in this study. Finally, $\sqrt{r^2 - a^2}/r \to 1$ as $a \to 0$, so $f_a(r) \to f_0(r) = 1 - 2M/r$ and Eq. (1) reduces directly to Schwarzschild in standard areal coordinates.

The first perturbation sector is a minimally coupled massless test scalar. Starting from

$$\Box\Phi = \frac{1}{\sqrt{-g}}\partial_\mu\left(\sqrt{-g}\, g^{\mu\nu}\partial_\nu\Phi\right) = 0 \tag{3}$$

and separating variables according to

$$\Phi(t, r, \theta, \phi) = e^{-\mathrm{i}\omega t} Y_{\ell m}(\theta, \phi)\frac{\Psi(r)}{r}, \tag{4}$$

with $\nabla^2_{S^2} Y_{\ell m} = -\ell(\ell + 1)Y_{\ell m}$, gives

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left[r^2 f_a(r)\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{\Psi}{r}\right)\right] + \left[\frac{\omega^2}{f_a(r)} - \frac{\ell(\ell+1)}{r^2}\right]\frac{\Psi}{r} = 0. \tag{5}$$

Introducing the tortoise coordinate $\mathrm{d}r_*/\mathrm{d}r = f_a^{-1}$ reduces this equation to the Schrödinger form. This coordinate stretches the horizon to $r_* \to -\infty$ and spatial infinity to $r_* \to +\infty$, putting the two radiative boundaries at the ends of a one-dimensional wave problem:

$$\frac{\mathrm{d}^2\Psi}{\mathrm{d}r_*^2} + \left[\omega^2 - V_{0\ell}(r;a)\right]\Psi = 0, \tag{6}$$

where the scalar effective potential is therefore

$$V_{0\ell}(r;a) = f_a(r)\left[\frac{\ell(\ell+1)}{r^2} + \frac{1}{r}\frac{\mathrm{d}f_a}{\mathrm{d}r}\right], \tag{7}$$

in agreement with the minimally coupled scalar equation used in Ref. [17]. QNM boundary conditions select solutions that are purely ingoing at the horizon and purely outgoing at spatial infinity. These conditions mean that energy leaves the exterior region through both ends and no wave is sent in from either boundary. They make ω complex and the radial problem non-self-adjoint.

Equation (7) also gives the useful physical picture. The $\ell(\ell+1)/r^2$ term is the angular-momentum barrier, while f_a'/r records the radial curvature of the geometry. Multiplication by the lapse makes the potential vanish at the horizon and at infinity, leaving a single barrier outside the hole. A QNM is a wave concentrated near this barrier and leaking through both sides. Moving and lowering the barrier therefore shifts the oscillation frequency, while changing its width and curvature changes the leakage rate and hence the damping.

The odd-parity sector requires an assumption beyond the spherical KS background. The original KS construction explicitly writes the full metric as a nonperturbative spherically symmetric part plus small propagating nonspherical modes and treats the latter classically at leading order [15]. Motivated by that separation, we represent the quantum-corrected spherical background through $G_{\mu\nu} = 8\pi T^{\mathrm{eff}}_{\mu\nu}$ and set the *gauge-invariant odd-parity current* of the effective source to zero. This is an additional inverse-Cowling (frozen-source) closure, not a constitutive law derived from the spherical KS effective action. It retains the KS spherical quantum backreaction but sets the odd part of $\delta T^{\mathrm{eff}}_{\mu\nu}$ to zero and therefore neglects independent nonspherical quantum-source excitations. The modes below are conditional on this explicitly defined model.

Physically, this closure lets the odd-parity geometry ring on the quantum-corrected spherical background while preventing the effective quantum source itself from carrying an independent odd-parity oscillation. It is the gravitational analogue of holding the material source fixed while perturbing the surrounding field. Gauge invariance ensures that the master amplitude is not a coordinate artifact; it does not determine whether the frozen-source assumption is the unique quantum dynamics.

Here *gauge* refers to the freedom to relabel coordinates without changing the physical spacetime. Individual perturbation components can change under such a relabelling, even though the underlying geometry does not. A gauge-invariant master variable is a combination in which those coordinate changes cancel. It therefore prevents a coordinate artifact from being mistaken for a physical wave, but it does not determine the dynamics of the effective quantum source.

To display the gauge invariance, write the background as $g_{ab}(y)\mathrm{d}y^a\mathrm{d}y^b + r^2(y)\gamma_{AB}\mathrm{d}z^A\mathrm{d}z^B$, where $a, b \in \{t, r\}$, and expand an odd perturbation as

$$\delta g_{aB} = h_a X_B^{\ell m}, \qquad \delta g_{AB} = h_2 X_{AB}^{\ell m}. \tag{8}$$

Under an odd infinitesimal diffeomorphism generated by $\xi_A = \Lambda X_A^{\ell m}$,

$$h_a \mapsto h_a - D_a\Lambda + \frac{2D_a r}{r}\Lambda, \qquad h_2 \mapsto h_2 - 2\Lambda. \tag{9}$$

With our harmonic normalization $h_2 = 2h$, the one-form

$$k_a = h_a - \frac{1}{2}D_a h_2 + \frac{D_a r}{r}h_2 \tag{10}$$

is therefore gauge invariant. This is Eq. (18) of Gundlach and Martínez-García [3], written in the Gerlach–Sengupta conventions of Ref. [2]. We choose the two-dimensional volume form so that $\epsilon^{tr} = +1$ in the static (t, r) chart and define, as in Eq. (68) of Ref. [3],

$$\Pi = \epsilon^{ab} D_b\left(\frac{k_a}{r^2}\right). \tag{11}$$

Writing the odd stress perturbation as $\delta T_{aB} = \delta t_a^{\rm odd} X_B$, its gauge-invariant source one-form is $L_a = \delta t_a^{\rm odd} - Q_{\rm bg} h_a$, where $Q_{\rm bg}$ denotes the coefficient called Q for the angular background stress tensor in the notation of Ref. [3]; this is their Eq. (19). Because it is constructed from the gauge-invariant matter combination, imposing $L_a = 0$ does not fix a coordinate gauge. The general sourced master equation, Eq. (69) of Ref. [3], is

$$D_a\left[\frac{1}{r^2}D^a(r^4\Pi)\right] - (\ell-1)(\ell+2)\Pi = -16\pi\epsilon^{ab}D_b L_a. \tag{12}$$

The associated conservation law is $D_a(r^2 L^a) = (\ell-1)(\ell+2)L$, Eq. (62) of the same reference. Our inverse-Cowling closure sets both odd source amplitudes $L_a = L = 0$, so the conservation equation and the right-hand side of Eq. (12) vanish identically. Equivalently, Karlovini's Eqs. (38), (39), and (46) give $(D^a D_a - U)\psi = \kappa r \epsilon^{ab} D_a(r^{-2} J_b)$, with the same source-free reduction [4].

For the static metric used here, the normalization $\Psi_{\rm ax} = r^3\Pi$ (up to an irrelevant constant) and $\mathrm{d}r_*/\mathrm{d}r = f_a^{-1}$ remove the first radial derivative from the source-free equation. After Fourier decomposition with $e^{-\mathrm{i}\omega t}$, it becomes

$$\frac{\mathrm{d}^2\Psi_{\rm ax}}{\mathrm{d}r_*^2} + \left[\omega^2 - V_{{\rm ax},\ell}(r;a)\right]\Psi_{\rm ax} = 0, \tag{13}$$

with the general anisotropic-source potential

$$V_{{\rm ax},\ell} = f_a\left[\frac{\ell(\ell+1)}{r^2} - \frac{6m(r)}{r^3} + 4\pi\left(\rho - p_r\right)\right], \qquad f_a = 1 - \frac{2m(r)}{r}. \tag{14}$$

This form is also obtained in the inverse-Cowling treatment of general static spherical backgrounds [Eqs. (14)–(17) of Ref. [5]].

For the KS metric, $g_{tt}g_{rr} = -1$. The background Einstein identities give

$$m(r) = \frac{r}{2}(1 - f_a) = M + \frac{r - \sqrt{r^2 - a^2}}{2}, \quad 4\pi\rho = \frac{m'}{r^2}, \quad p_r = -\rho. \tag{15}$$

Substitution into Eq. (14) yields the gauge-invariant inverse-Cowling KS axial potential used in all calculations:

$$V_{{\rm ax},\ell}(r;a) = f_a(r)\left[\frac{\ell(\ell+1)}{r^2} + \frac{2(f_a - 1)}{r^2} - \frac{f_a'}{r}\right]. \tag{16}$$

Equation (16) is the gauge-invariant master potential obtained under the inverse-Cowling closure $L_a = L = 0$, not a unique prediction of the nonspherical KS quantum theory; the associated axial frequencies should therefore be interpreted as conditional model results. At $a = 0$, $m = M$, $\rho = p_r = 0$, and Eq. (16) becomes the Schwarzschild Regge–Wheeler potential [1]. For comparison, define the lapse-substitution potential

$$V_{\mathrm{lapse},\ell}(r;a) = f_a(r)\left[\frac{\ell(\ell+1)}{r^2} - \frac{6M}{r^3}\right]. \tag{17}$$

For $a > 0$, the inverse-Cowling potential differs from this comparison potential by

$$\Delta V_{\mathrm{ax}} = \frac{f_a}{r^3}\left[2\left(\sqrt{r^2-a^2} - r\right) - \frac{a^2}{\sqrt{r^2-a^2}}\right], \tag{18}$$

so the two models are not equivalent. The same effective-source potential is also used in the published high-precision Chebyshev calculation of Batic, Dutykh, and Sukaiti [6]; Section 3 compares overlapping modes directly. We use this potential to compute the low-lying spectrum with two numerically distinct radial discretizations, monitoring residuals, backward error, continuation in a/M, and local pseudospectral sensitivity. Gauge invariance removes coordinate artifacts from the metric master variable; it does not make the inverse-Cowling closure unique. A nonspherical quantum-source action could instead generate a nonzero right-hand side or additional couplings in Eq. (12) [2, 3, 4].

2.2 Numerical method

We determine the same physical resonances using two numerically distinct frequency-domain representations. The Chebyshev calculation describes the radial wave globally on a compact grid and produces candidate ringing frequencies. The Frobenius calculation instead expands the wave locally and enforces the horizon and infinity conditions through a continued fraction. Agreement between them shows that a reported mode is tied to the wave equation and its radiative boundaries rather than to one discretization. A time-domain evolution of the scalar equation provides an additional, lower-precision check on the direction and scale of the frequency shift, following earlier KS studies [17]. The quantitative spectrum comes from the two frequency-domain calculations [19, 28]; the matrix-pencil fit to the waveform is only an orientation diagnostic.

2.2.1 Baseline time-domain diagnostic

The *time domain* means solving directly for the waveform as a function of time and then fitting its ringdown. It answers the intuitive question "what would the decaying signal look like after an initial pulse?" This method is especially good at recovering the least-damped mode because that mode survives longest, but several damped sinusoids can be difficult to separate in a short or noisy time interval. We therefore use it to check the sign and approximate size of the deformation shift, not to set the final digits.

The baseline evolves the radial wave equation in tortoise coordinate $x = r_*$,

$$\frac{\partial^2\Psi}{\partial t^2} - \frac{\partial^2\Psi}{\partial x^2} + V_{0\ell}(r(x);a)\Psi = 0. \tag{19}$$

For orientation, Eq. (19) is evolved on $x/M \in [-250, 500]$, with $\Delta x = 0.1M$, $\Delta t = 0.05M$, $t_{\max} = 320M$, and observation point $x_{\mathrm{obs}} = 20M$. A Gaussian pulse $\Psi(0,x) = \exp[-(x/4M)^2]$

is evolved with first-order outgoing boundary conditions, as in standard time-domain QNM extractions [26, 11, 17]. The interval $60M \le t \le 160M$ is fitted by

$$\Psi_{\rm fit}(t) = Ae^{-\alpha(t-t_0)}\cos(\omega_R t + \phi) + C_0, \qquad \omega_{\rm fit} = \omega_R - \mathrm{i}\alpha.$$

A rank-four matrix-pencil reduction is also applied to the same waveform [27]. It recovers the expected ringdown scale and deformation trend, but no frequency reported as a spectral result is inferred from this fit. Here a *matrix pencil* is a signal-processing construction that represents the sampled waveform as a sum of damped exponentials; its eigenvalues encode their frequencies and decay rates. It should not be confused with the generalized matrix pencil obtained below from the differential equation.

2.2.2 Chebyshev spectral discretization

Chebyshev polynomials are oscillatory global functions on $[-1, 1]$ whose special nodes are dense near the endpoints. Interpolating at these nodes reduces large endpoint errors and gives differentiation matrices that differentiate the interpolating polynomial. *Collocation* means that we require the differential equation to hold at each node. Increasing N raises the polynomial resolution and supplies a direct convergence test.

The direct spectral method compactifies the horizon-to-infinity domain with

$$s = 1 - \frac{r_h}{r}, \qquad x = 2s - 1, \qquad x \in [-1, 1],$$

where $r_h = \sqrt{4M^2 + a^2}$. Chebyshev-Gauss collocation points

$$x_j = \cos\left(\frac{\pi(j + 1/2)}{N}\right), \qquad j = 0, \ldots, N-1,$$

are mapped to $s_j = (x_j + 1)/2$, excluding the two singular endpoints. Barycentric differentiation matrices D and D^2 approximate ∂_s and ∂_s^2, following the usual pseudospectral treatment of QNM boundary-value problems [28].

The QNM boundary behavior is imposed analytically by factoring

$$\Psi(r) = e^{\mathrm{i}\omega r}\, r^{2\mathrm{i}M\omega}\, s^{-\mathrm{i}\omega/f_a'(r_h)}\, u(s). \tag{20}$$

The factorization in Eq. (20) builds the outgoing behavior at infinity and the ingoing behavior at the horizon into the unknown, leaving $u(s)$ regular on the collocation grid [19, 28]. Substitution into Eq. (6) gives the quadratic eigenvalue problem

$$P_N(\omega)\boldsymbol{u} \equiv \left(A_0 + \omega A_1 + \omega^2 A_2\right)\boldsymbol{u} = 0. \tag{21}$$

The matrices A_0, A_1, A_2 are built directly from $f_a(r)$, $f_a'(r)$, the selected potential $V_{0\ell}$ or $V_{\mathrm{ax},\ell}$, and the chain-rule factors relating r and s.

2.2.3 Residual test and mode selection

The eigensolver is deliberately treated as a candidate generator. The reason is that an algebraic eigenvalue of a finite matrix need not approximate a QNM of the original differential equation. The residual asks whether some normalized discrete wavefunction actually satisfies that equation at the trial frequency. A singular value is a non-negative measure of how strongly a matrix stretches vectors; $\sigma_{\min}$ is the weakest stretching direction. It vanishes exactly when the matrix has a nonzero null vector.

For a trial frequency we evaluate the residual map

$$\omega \mapsto \sigma_{\min}(P_N(\omega))\,,$$

where $P_N(\omega)$ is the finite-dimensional Chebyshev representation of the factored radial equation. A true finite-dimensional eigenfrequency drives this singular value to zero; in floating-point arithmetic a well-resolved mode appears as a sharp local minimum. This is the same singular-value viewpoint used in nonlinear eigenvalue analysis and black-hole pseudospectra [34, 32, 33].

The quadratic pencil first supplies possible roots in a damped frequency window. We then minimize $\sigma_{\min}[P_N(\omega)]$ locally, compare the mode across several values of N, follow it continuously in a/M, and solve the corresponding Frobenius recurrence. The tables report the frequency together with the raw residual, a scale-aware polynomial backward error, and the branch classification. In this way the generalized eigensolve locates the resonance, while convergence and the continued fraction determine how much confidence can be attached to it [19, 28].

2.2.4 *Generalized eigenproblem and residual*

The equation is *quadratic* because it contains 1, ω, and ω^2. Linearization doubles the state vector and rewrites it as a *generalized eigenproblem* $Az = \omega Bz$. The pair (A, B) is often called a matrix pencil. Its generalized eigenvalues are easy to compute, but the doubling and the different coefficient scales can amplify roundoff; this is why the resulting frequencies are candidates rather than accepted modes.

Equation (21) is linearized as

$$\begin{pmatrix} 0 & I \\ -A_0 & -A_1 \end{pmatrix} \begin{pmatrix} \boldsymbol{u} \\ \omega \boldsymbol{u} \end{pmatrix} = \omega \begin{pmatrix} I & 0 \\ 0 & A_2 \end{pmatrix} \begin{pmatrix} \boldsymbol{u} \\ \omega \boldsymbol{u} \end{pmatrix}, \tag{22}$$

We solve Eq. (22) with `scipy.linalg.eig`. Its roots are initial guesses rather than final frequencies, since polynomial linearizations can magnify scaling and conditioning problems. The scalar $\ell = 2$ fundamental is identified from the Schwarzschild spectrum, whereas overtones are followed by continuity of both frequency and eigenfunction and are checked separately with the continued fraction [19, 28].

2.2.5 *Root filtering, refinement, and branch assignment*

Equilibration rescales rows and columns so that very large and very small coefficients enter the eigensolve on comparable numerical scales. In exact arithmetic this does not change a finite eigenvalue; in floating-point arithmetic it can reduce avoidable loss of precision. Filtering then removes frequencies outside the damped physical window and merges duplicate numerical representatives. Refinement varies the real and imaginary parts of ω to find the bottom of the local residual basin. Branch assignment is a separate step: it asks which refined root is the continuation of a particular (ℓ, n) mode rather than merely which root is closest at one parameter value.

The selection rules are fixed before the spectrum is computed. Before solving the pencil, each row of the pair of linearized matrices is divided by the larger of its two row infinity norms; the same operation is then applied columnwise after row scaling. Scaling factors are clipped to $[10^{-100}, 10^{100}]$, and the right generalized eigenvectors are mapped back to the unscaled coordinates. Infinite and non-finite roots are removed. The physical search window is $0.05 < \operatorname{Re}\omega < 2$ and $-3 < \operatorname{Im}\omega < -0.02$. Roots separated by at most 10^{-7} are treated as one numerical cluster, retaining the representative with the smaller raw-polynomial singular-value residual. At most the 20 candidates closest to the current branch target enter the detailed scoring stage.

For a previously tracked branch, a candidate must lie within an absolute complex-frequency distance 0.20 of the preceding target. Competing matches minimize

$$S = d_{\text{target}} + 0.45 d_{\text{cont}} + 0.35(1 - |\langle u_{j-1}, u_j\rangle|) + 10^{-3}\min(\sigma_j/\sigma_{\text{best}}, 100),$$

where, for the shortlisted candidate set C,

$$d_{\text{target}} = \frac{|\omega_j - \omega_{\text{target}}|}{\max(|\omega_{\text{target}}|, 0.1)}, \qquad d_{\text{cont}} = \frac{|\omega_j - \omega_{j-1}|}{\max(|\omega_{j-1}|, 0.1)}, \qquad \sigma_{\text{best}} = \min_{k\in C}\sigma_k.$$

Here $\sigma_j = \sigma_{\min}[P_N(\omega_j)]$ is evaluated on the raw, unequilibrated polynomial matrices at the unrefined generalized-eigenvalue candidate. Candidate frequencies and right eigenvectors are generated from the row- and column-equilibrated linearized pencil; before overlap evaluation, the right eigenvectors are mapped back by the column-scaling matrix as $x = D_r y$. The extracted mode shapes are normalized, and the previous shape is barycentrically interpolated to the new grid. The absolute value in $|\langle u_{j-1}, u_j\rangle|$ removes the arbitrary complex phase of either eigenvector. The selected raw root is refined within a rectangular radius 0.02 in each complex-frequency component by L-BFGS-B minimization of $\sigma^2_{\min}[P_N(\omega)]$, with at most 500 iterations, `ftol`$= 10^{-18}$, and `gtol`$= 10^{-12}$.

The catalogue labels $n = 0, 1, 2$ begin from the stated Schwarzschild seeds; after each deformation value, the corresponding continued-fraction roots become the three targets for the next value, and already assigned roots are excluded within 10^{-7}. A catalogue row is retained only when its raw-polynomial backward error is below 10^{-8} and its spectral–continued-fraction relative difference is below 10^{-4}. These thresholds remove clearly unresolved roots, but they do not by themselves establish the physical reliability of a branch. In particular, the $n = 2$ modes remain exploratory because their convergence is visibly less uniform.

For completeness, the implemented sequence is: (1) equilibrate and solve the quadratic pencil; (2) remove non-finite roots and roots outside the damped window; (3) cluster at 10^{-7}; (4) enforce the 0.20 continuation gate and score competing matches; (5) refine the selected Chebyshev root by bounded singular-value minimization; (6) apply the backward-error and spectral–continued-fraction gates; and (7) propagate the accepted continued-fraction root as the next deformation target. Thus the continued-fraction calculation is collocation-independent, although it shares the perturbation equation and endpoint factorization.

The direct spectral residual is

$$R_N(\omega) = P_N(\omega)^\dagger P_N(\omega). \tag{23}$$

Equation (23) is Hermitian and positive semidefinite up to roundoff. We quote $\sigma_{\min}[P_N(\omega)]$ and the scale-aware polynomial backward error, but neither number is meaningful in isolation: a small residual can accompany a poorly conditioned or nonconvergent discretized root [34]. Spectral-size convergence and the continued-fraction result therefore carry equal weight in the mode assignment.

2.2.6 *Leaver-style validation*

As a collocation-independent frequency-domain calculation, we also solve a Frobenius recurrence in the manner of Leaver [19, 20]. A Frobenius series is a power series multiplied by the known singular behavior at an endpoint. Substitution converts the differential equation into a recurrence among series coefficients. For a generic frequency the recurrence contains a solution that grows and violates the required boundary behavior; the continued fraction is the condition that selects the minimal, physically admissible coefficient sequence. It uses the same compact coordinate $s = 1 - r_h/r$ and the same physical endpoint behavior, but no Chebyshev grid, matrix-pencil roots, or singular-value minimization. We write

$$\Psi(r) = e^{\mathrm{i}\omega r} r^{2\mathrm{i}M\omega} s^{-\mathrm{i}\omega/f'(r_h)} u(s),$$

and expand the regular factor as a Frobenius series,

$$u(s) = \sum_{n=0}^{\infty} a_n s^n.$$

Substitution into the factored radial equation gives

$$B_2(s)u'' + B_1(s)u' + B_0(s)u = 0,$$

where the $B_j(s)$ are generated by formal power-series algebra from the KS metric functions. The Taylor-truncated Frobenius equations define a multi-term recurrence for the coefficients a_n. Gaussian elimination reduces this recurrence to the three-term form

$$\alpha_n a_{n+1} + \beta_n a_n + \gamma_n a_{n-1} = 0,$$

and the minimal-solution condition is imposed through Leaver's continued fraction,

$$\beta_0 - \cfrac{\alpha_0\gamma_1}{\beta_1 - \cfrac{\alpha_1\gamma_2}{\beta_2 - \cdots}} = 0. \tag{24}$$

The reported calculation expands each B_j through Taylor order 96 and evaluates the reduced recurrence to continued-fraction depth 240. Roots of Eq. (24) are obtained in IEEE-754 double precision by applying the MINPACK hybrid solver exposed by `scipy.optimize.root` to the real and imaginary residual components. The root tolerance is 10^{-11}, the maximum number of residual-function evaluations is 1000, and the accepted residual is required to be below 10^{-7} for the scalar validation calculation. The initial guesses are the branch-tracked $N = 32$ Chebyshev candidates. At $a/M = 0$, those candidates are selected as the nearest distinct roots to the 18 literature targets listed in Table 11 of Appendix A; no manual or continued-fraction Schwarzschild seed is used. Branch identity is maintained by continuation over $a/M = 0, 0.2, 0.5, 1$ and by keeping the fundamental and first-overtone targets distinct. The catalogue uses the same settings, with a relaxed 10^{-4} residual acceptance threshold for the most delicate second-overtone recurrence reductions. This validation layer is collocation-independent: it uses no Chebyshev grid, matrix-pencil data, or residual minimization, while sharing the same perturbation equation and endpoint factorization.

3 Results

Before discussing numerical accuracy, it is useful to state the physical effect visible in the scalar wave equation. At fixed M, increasing the deformation from $a/M = 0$ to 1 moves the horizon from $r_h/M = 2$ to $\sqrt{5}$. For the $\ell = 2$ scalar barrier, the peak moves from $r/M = 2.9517$ to 3.2520, its height falls by 13.54%, and the magnitude of its tortoise-coordinate curvature falls by 17.77%. The leading WKB estimate $\mathrm{Re}\,\omega \sim \sqrt{V_{\text{peak}}}$ therefore predicts a 7.01% decrease in oscillation frequency, close to the fully computed 7.29% shift. The damping rate depends on both the barrier curvature and the global outgoing-wave solution, so it need not fall by the same fraction; the numerical mode gives a smaller 2.59% decrease in damping magnitude.

In time-domain language, the deformed scalar signal would have a lower pitch and a slightly more slowly decaying envelope. Because the pitch falls more than the damping rate, the signal nevertheless completes fewer oscillations within one damping time: its quality factor decreases by 4.83%. The $\ell = 3$ and 4 barriers show the same behavior, so this is a systematic low-lying scalar effect rather than an isolated quadrupole feature. The following validation establishes that these percent-level changes are much larger than the numerical uncertainty.

3.1 Schwarzschild validation

As an external benchmark we use the spin-zero Schwarzschild fundamental with $\ell = 2$, overtone index $n = 0$, mass normalization $M = 1$, time dependence $e^{-\mathrm{i}\omega t}$, and therefore $\mathrm{Im}\,\omega < 0$ for a damped mode. In the neutral Schwarzschild limit $q = Q = 0$, Cavalcante and da Cunha report the spin-zero $\ell = 2$ entry in Table I on page 6 from an isomonodromic Painlevé calculation and reproduce it in the adjacent column with a continued-fraction method [35]. After matching our $e^{-\mathrm{i}\omega t}$ convention, the value is

$$M\omega_{\text{ref}} = 0.483643872211 - 0.096758775978\,\mathrm{i}.$$

Throughout the numerical comparisons we use

$$\Delta_{\text{rel}}(\omega_1, \omega_2) = \frac{|\omega_1 - \omega_2|}{|\omega_2|}.$$

Here ω_2 is the reference value. At $N = 96$, the direct spectral calculation gives

$$M\omega_{\text{spec}} = 0.483643872189 - 0.096758775961\,\mathrm{i},$$

with relative difference 5.60×10^{-11} from that external value. This is substantially sharper than the baseline time-domain fit, $0.483642480 - 0.096758919\,\mathrm{i}$, and the waveform matrix-pencil diagnostic, $0.483653530 - 0.096763421\,\mathrm{i}$.

The Leaver-style solver gives

$$M\omega_{\text{Leaver}} = 0.483643872211 - 0.096758775978\,\mathrm{i}$$

for the Schwarzschild fundamental mode, with relative difference 8.24×10^{-13} from the same external value. Such accuracy is routine for mature Schwarzschild solvers [19, 35]; its role here is to verify the conventions, endpoint factors, and branch identification in both of our frequency-domain implementations.

Figure 1 shows how the Chebyshev roots approach an order-128, depth-320 continued-fraction value. The fundamental settles onto a double-precision plateau. The first overtone behaves differently: agreement is best near $N = 32$–48 and worsens as the differentiation matrices become more ill-conditioned at larger N, a familiar limitation of high-order collocation in finite precision [28]. We therefore describe these modes as cross-validated low-lying first overtones at $N = 32$, rather than interpreting the largest available grid as automatically the most accurate.

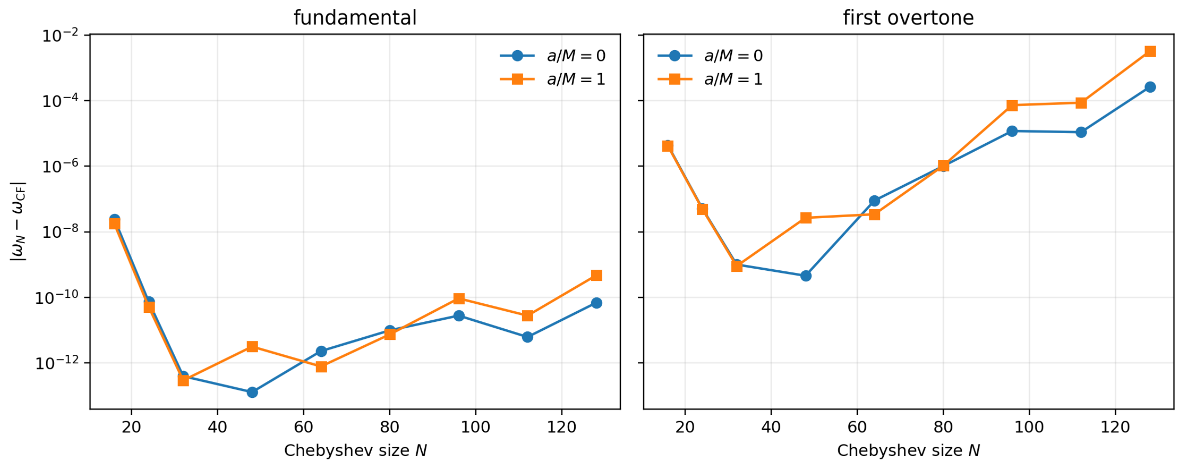

Figure 1 Chebyshev error relative to the order-128, depth-320 continued-fraction root for scalar $\ell = 2$ fundamentals (left) and first overtones (right), including $N = 128$. The overtone panel displays the high-N deterioration directly.

As an upper-grid check, we also recomputed the scalar $\ell = 2, n = 0$ branch at $N = 128$ for $a/M = 0$ and 1. The shifts from $N = 96$ are 6.95×10^{-11} and 4.64×10^{-10}, respectively, far below the scale of any deformation trend reported below. The small nonmonotonic motion between $N = 96$ and 128 is consistent with roundoff amplification in the differentiation matrices and the linearized pencil [28]. We use $N = 96$ for the fundamental catalogue and take the collocation-independent continued fraction as the external numerical anchor.

3.2 Leaver continued-fraction comparison

Table 1 summarizes the Leaver-style continued-fraction validation against the targeted Chebyshev spectral rows used for the scalar comparison.

a/M	mode	$\mathrm{Re}(M\omega_{\mathrm{Leaver}})$	$\mathrm{Im}(M\omega_{\mathrm{Leaver}})$	$\Delta_{\mathrm{rel}}(\mathrm{spec}, \mathrm{Leaver})$
0.0	fundamental	0.48364387	-0.09675878	7.914×10^{-13}
0.0	first overtone	0.46385058	-0.29560394	1.814×10^{-9}
0.2	fundamental	0.48203369	-0.09665386	6.084×10^{-13}
0.2	first overtone	0.46216559	-0.29531130	1.788×10^{-9}
0.5	fundamental	0.47388693	-0.09610918	6.219×10^{-13}
0.5	first overtone	0.45364275	-0.29379018	1.808×10^{-9}
1.0	fundamental	0.44836241	-0.09424818	6.341×10^{-13}
1.0	first overtone	0.42697149	-0.28857291	1.747×10^{-9}

Table 1 Leaver-style continued-fraction validation against targeted Chebyshev spectral results at $N = 32$. Frequency columns are rounded for display; the discrepancy column is computed from full-precision internal values. The Leaver solver avoids Chebyshev collocation, matrix-pencil data, and residual minimization, while sharing the same perturbation equation and endpoint factorization. The stricter scalar comparison shown here fails if the relative difference exceeds 10^{-6}; the looser general catalogue gate of 10^{-4} exists only to accommodate the exploratory $n = 2$ tier. The largest difference in this table is 1.814×10^{-9}.

Table 2 isolates continued-fraction truncation from the spectral comparison. The fundamental modes stabilize by depth 90 at the shown precision. The deformed first overtone is more sensitive, but its depth-240 and depth-320 values differ by only 2.05×10^{-12}. This is consistent with treating the scalar $n = 1$ modes as cross-validated low-lying first overtones at $N = 32$ while retaining a stronger precision assignment for the fundamentals.

case	CF depth	$\mathrm{Re}(M\omega)$	$\mathrm{Im}(M\omega)$	$\lvert\omega - \omega_{320}\rvert$
Schwarzschild $n = 0$	60	0.483643872211	-0.096758775978	2.68×10^{-13}
	90	0.483643872211	-0.096758775978	4.89×10^{-16}
	120	0.483643872211	-0.096758775978	0
	240	0.483643872211	-0.096758775978	0
	320	0.483643872211	-0.096758775978	0
KS $a/M = 1, n = 0$	60	0.448362409002	-0.094248179159	1.76×10^{-13}
	90	0.448362409002	-0.094248179159	1.35×10^{-15}
	120	0.448362409002	-0.094248179159	3.93×10^{-16}
	240	0.448362409002	-0.094248179159	5.55×10^{-17}
	320	0.448362409002	-0.094248179159	0
KS $a/M = 1, n = 1$	60	0.426971489725	-0.288572913307	2.09×10^{-10}
	90	0.426971489574	-0.288572913453	1.12×10^{-12}
	120	0.426971489576	-0.288572913449	2.84×10^{-12}
	240	0.426971489574	-0.288572913453	2.05×10^{-12}
	320	0.426971489574	-0.288572913452	0

Table 2 Continued-fraction depth check at fixed Taylor order 96. Each solve uses the branch-tracked $N = 32$ Chebyshev value only as an initial guess; the last column compares with the depth-320 root. The table is generated by `scripts/analyze_leaver_convergence.py`.

Taylor-order convergence was tested separately at fixed continued-fraction depth 320. Table 3 reports the complex-frequency difference from order 128. At the production order 96 the largest difference among these cases is 4.32×10^{-12}.

case	64	80	96	112	128
scalar $\ell = 2, n = 0, a/M = 1$	2.90×10^{-13}	1.60×10^{-14}	2.50×10^{-15}	0	0
scalar $\ell = 2, n = 1, a/M = 1$	1.20×10^{-9}	4.32×10^{-12}	4.32×10^{-12}	3.30×10^{-13}	0
axial $\ell = 2, n = 0, a/M = 1$	3.09×10^{-12}	1.67×10^{-13}	6.13×10^{-14}	3.40×10^{-15}	0

Table 3 Absolute difference $|\omega_p - \omega_{128}|$ under one-at-a-time variation of the Frobenius Taylor order p, with continued-fraction depth 320 fixed. Generated by `scripts/analyze_solver_convergence.py`.

3.3 Gauge-invariant odd-parity gravitational sector

We next apply the same Chebyshev and continued-fraction calculations to the gauge-invariant inverse-Cowling axial equation for $\ell = 2, 3, 4$ and low-lying indices $n = 0, 1, 2$. Together with the scalar rows, the working catalogue contains 72 entries over $a/M = 0, 0.2, 0.5, 1.0$. The axial entries are conditional inverse-Cowling modes of the explicitly stated closure. At $a = 0$, the effective source vanishes and the model reduces exactly to the Schwarzschild Regge–Wheeler equation [1, 2, 4]. Table 4 lists the resulting continued-fraction frequencies. External high-precision comparisons are reported separately below.

ℓ	n	$\mathrm{Re}(M\omega_{\mathrm{Leaver}})$	$\mathrm{Im}(M\omega_{\mathrm{Leaver}})$
2	0	0.37367168	-0.08896232
2	1	0.34671100	-0.27391488
2	2	0.30105346	-0.47827698
3	0	0.59944329	-0.09270305
3	1	0.58264380	-0.28129811
3	2	0.55168490	-0.47909275
4	0	0.80917838	-0.09416396
4	1	0.79663153	-0.28433435
4	2	0.77270953	-0.47990818

Table 4 Schwarzschild Regge–Wheeler frequencies obtained by the continued-fraction solver. At $a = 0$, the effective-source model reduces exactly to the Schwarzschild Regge–Wheeler equation. Because available tabulations have uneven numbers of significant figures, their rounded entries are used only as branch-identification targets; the quantitative accuracy check against a named high-precision calculation is reported separately below.

Representative gauge-invariant inverse-Cowling axial $\ell = 2$ trajectories over the deformation grid are shown in Figure 2.

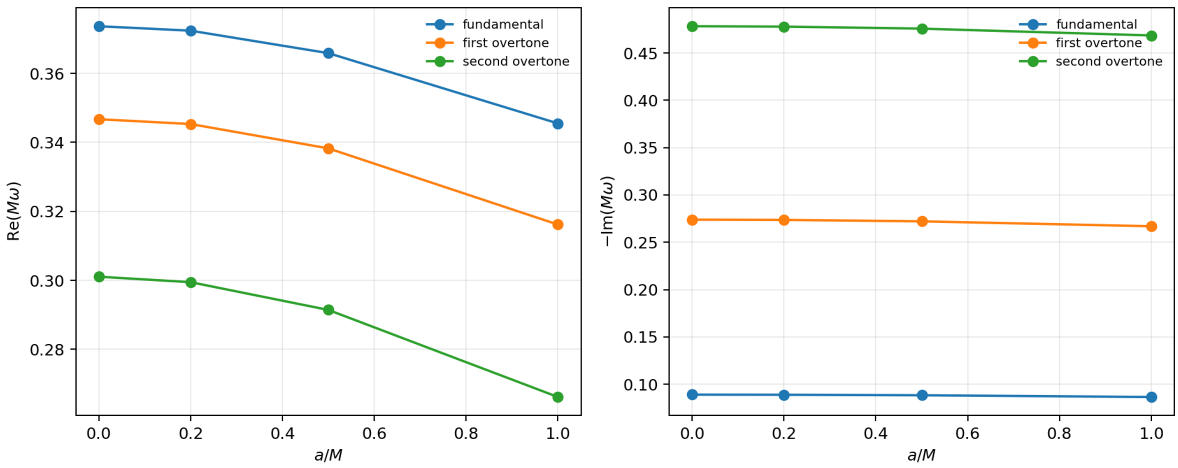

Figure 2 Representative $\ell = 2$ conditional odd-parity mode trajectories of the gauge-invariant inverse-Cowling effective-source model versus a/M. They are not predictions of an unspecified quantum nonspherical completion.

Figure 3 compares the inverse-Cowling potential with the lapse-substitution potential in Eq. (17). At $a/M = 1$, the maximum absolute correction is 5.77% of the new barrier height; over the region where the new potential exceeds 10^{-3} of its peak, the maximum pointwise fractional correction is 15.1%. These definitions avoid a spurious relative divergence at the horizon and infinity, where both potentials vanish.

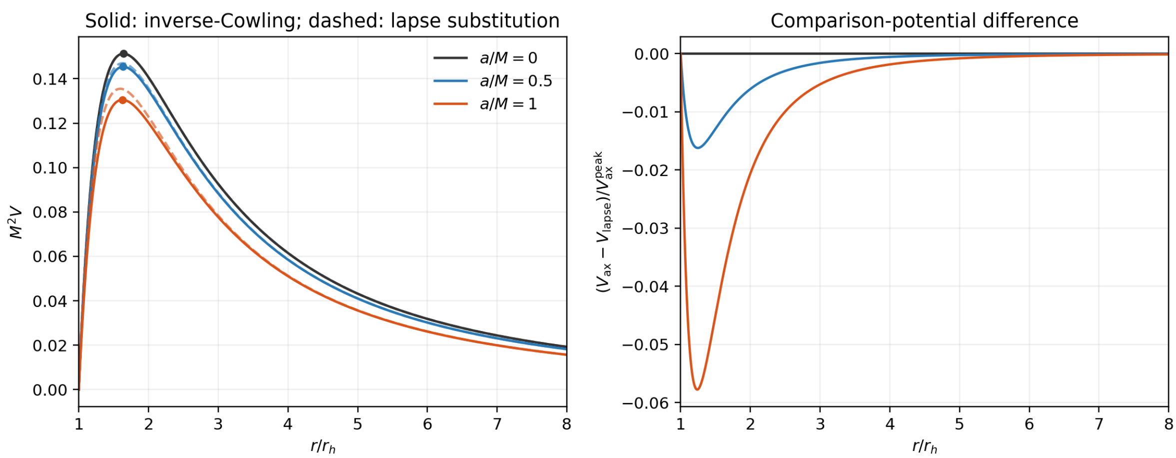

Figure 3 Left: inverse-Cowling axial potential (solid) and the explicitly defined lapse-substitution comparison potential (dashed) for $\ell = 2$; circles mark the inverse-Cowling potential peaks and the vertical line marks $r = r_h$. Right: their difference normalized by the new barrier height.

The same potential and mass normalization are used by the published high-precision Chebyshev calculation of Batic, Dutykh, and Sukaiti [6]. We compared all 18 overlapping $a/M = 0.5, 1$, $\ell = 2, 3, 4$, $n = 0, 1, 2$ entries. The largest relative difference is 1.98×10^{-13} for fundamentals, 3.80×10^{-10} for first overtones, and 1.62×10^{-5} for second overtones. The last occurs at $a/M = 1, \ell = 2, n = 2$ and reinforces the exploratory status of that tier. The comparison, parameter conversion check, public repository commit, and local source commit are recorded by `scripts/audit_axial_model.py`. The external numerical audit used repository commit `f53435ebdb8d1124d13bee75fa54972ac1613a03` [7].

As a stability diagnostic, the potential was minimized numerically outside the horizon for every sampled a/M and $\ell = 2, 3, 4$. It remained non-negative and approached zero only at the two boundaries. The finite-spectrum search was applied before the ordinary damped-mode window: all finite roots satisfying $\operatorname{Im}\omega > 10^{-8}$ and $|\omega| < 5$ were retained at $N = 32, 48, 64$, and persistence required agreement within 10^{-4} at all three sizes. Any persistent candidate was to be locally residual-refined within radius 0.02 at $N = 64$ and accepted only if its polynomial backward error was below 10^{-8}. No raw positive-imaginary root occurred, so the refinement stage was not invoked and no growing candidate survived. Positivity supplies the usual sufficient mode-stability argument for this source-free one-dimensional problem. For a hypothetical mode with $\operatorname{Im}\omega > 0$, the QNM boundary conditions make the radial solution decay at both asymptotic ends; multiplying the source-free master equation by $\Psi^*_{\rm ax}$, integrating, and using the standard positive-energy identity excludes a growing mode when $V_{\rm ax} \geq 0$ [5]. This statement applies only to the inverse-Cowling model and does not establish stability under other effective-source closures; the finite root search is an additional numerical check.

Across the low-lying catalogue, the largest spectral–continued-fraction difference is 1.202×10^{-5}, for the axial $\ell = 2, n = 2$ branch at $a/M = 0.5$. The axial fundamentals are much more stable numerically. We keep the $n = 1$ and $n = 2$ axial values because they show how the branches move, but all axial modes inherit the inverse-Cowling assumption and the $n = 2$ entries carry the additional overtone uncertainty familiar from KS studies [18].

3.4 Catalogue-level deformation trends

To isolate the effect of the KS deformation, we compare each branch with its own Schwarzschild value at fixed M. Both the real frequency and the damping magnitude decrease monotonically on the sampled grid for every scalar branch. The inverse-Cowling axial branches follow the same trend, although their physical interpretation remains conditional on the frozen-source assumption.

At the upper endpoint of the sampled interval, $a/M = 1$, the scalar $\ell = 2$ fundamental shifts are -7.29% in $\operatorname{Re}(M\omega)$, -2.59% in the damping magnitude $-\operatorname{Im}(M\omega)$, and 7.17% in the complex-plane displacement $|\Delta\omega|/|\omega(a = 0)|$. These are distinct quantities and are not referred to interchangeably below. Among the scalar fundamentals, the $\ell = 4$ branch has the largest complex-plane displacement, 7.23%, while its real-part shift is −7.27%. The scalar $\ell = 2$ second overtone has a larger real-part shift, −9.16%, but is an exploratory branch diagnostic. The $\ell = 2$ trends are plotted in Figure 4.

Within the closure-dependent axial fundamentals, the largest complex-plane displacement is 7.37% for $\ell = 2$, with shifts of −7.54% in the real part and −2.80% in damping magnitude. This value is reported separately because it depends on the frozen-source assumption, whereas the scalar comparison above does not.

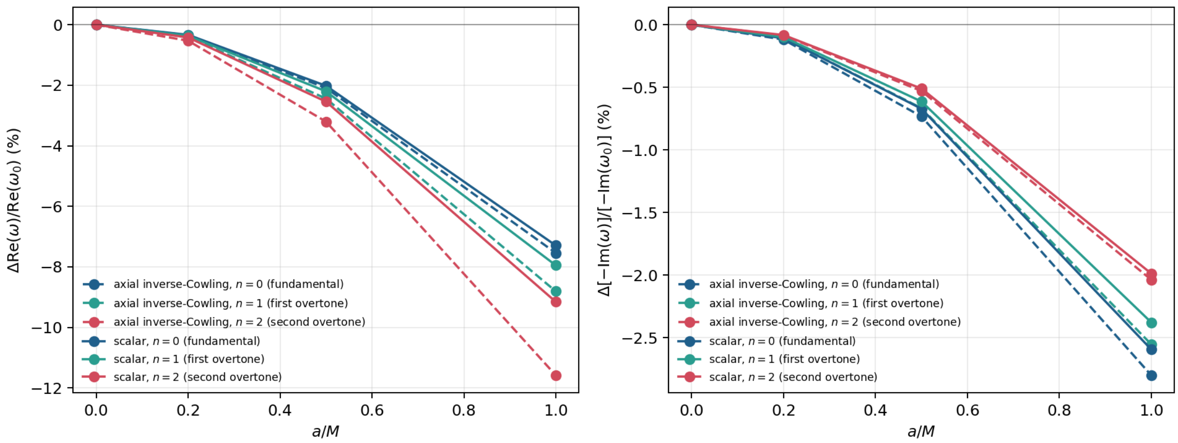

Figure 4 Schwarzschild-relative trends over the sampled interval for the $\ell = 2$ branches. Solid curves show scalar modes and dashed curves show the gauge-invariant inverse-Cowling axial modes. The latter are conditional on the closure stated in Section 2.1.

The mild increase of the scalar fundamental displacement from $\ell = 2$ to $\ell = 4$ is already visible in the effective-potential barrier. Table 5 reports the peak analysis at $a/M = 1$. The leading WKB proxy $\mathrm{Re}\,\omega \sim \sqrt{V_{\mathrm{peak}}}$ changes by −7.01%, −7.13%, and −7.18% for $\ell = 2, 3, 4$, respectively, mirroring the modest multipole dependence of the computed fundamentals. The curvature change also grows slightly with ℓ, consistent with a weak multipole-dependence in the damping. This is precisely the level of intuition expected from leading WKB theory [23, 24]; the data do not suggest a separate $\ell = 4$ mechanism.

ℓ	r_{peak}/M	ΔV_{peak}	$\Delta\sqrt{V_{\mathrm{peak}}}$	$\Delta\lvert V''_{*,\mathrm{peak}}\rvert$
2	3.25198574	−13.5350%	−7.0134%	−17.7698%
3	3.28015848	−13.7535%	−7.1310%	−18.1334%
4	3.29264164	−13.8440%	−7.1797%	−18.2801%

Table 5 Scalar effective-potential peak changes at $a/M = 1$, relative to the Schwarzschild peak at the same ℓ. Primes denote tortoise-coordinate derivatives. These diagnostics are generated by `scripts/analyze_scalar_potential_peaks.py` and provide only WKB intuition; the QNM frequencies come from the Chebyshev and continued-fraction calculations.

3.5 Spectroscopy diagnostics

At fixed mass, the shift has a straightforward spectroscopic meaning. The decrease in $\mathrm{Re}(M\omega)$ lowers the oscillation frequency, while the decrease in $-\mathrm{Im}(M\omega)$ lengthens the damping time. Following the standard ringdown convention [10, 13], we define

$$Q = \frac{\mathrm{Re}\,\omega}{2[-\mathrm{Im}\,\omega]}.$$

If $\tau = 1/[-\mathrm{Im}\,\omega]$ is the time for the mode amplitude to fall by a factor of e, then the oscillation advances through $2Q$ radians, or Q/π cycles, during that time. Thus Q measures how many oscillations remain visible before damping substantially reduces the signal. For the scalar

$\ell = 2$ fundamental branch, the damping magnitude decreases by 2.59%, corresponding to an approximate 2.7% increase in damping time. Because the oscillation frequency decreases by a larger fraction, Q decreases by 4.83%: the scalar perturbation decays slightly more slowly, but it completes fewer oscillations per damping time.

Dimensionless mode ratios are useful because a common rescaling of the black-hole mass cannot absorb them, which is why they play a central role in ringdown spectroscopy [13, 14]. For the scalar $\ell = 2$ branch at $a/M = 1$, the complex ratio ω_0/ω_1 changes from $0.836060 + 0.324207\,\mathrm{i}$ to $0.823241 + 0.335659\,\mathrm{i}$, a 1.92% displacement in the complex-ratio plane. The corresponding ω_0/ω_2 ratio moves by 3.56%. The branch-wise quantity $\mathrm{Re}(\omega)/[-\mathrm{Im}(\omega)] = 2Q$ changes by -4.83%, -5.71%, and -7.31% for $n = 0, 1, 2$, respectively. The $n = 2$ number is included only as an exploratory diagnostic. Figure 5 collects these ratios and the fundamental $Q(a/M)$ curve.

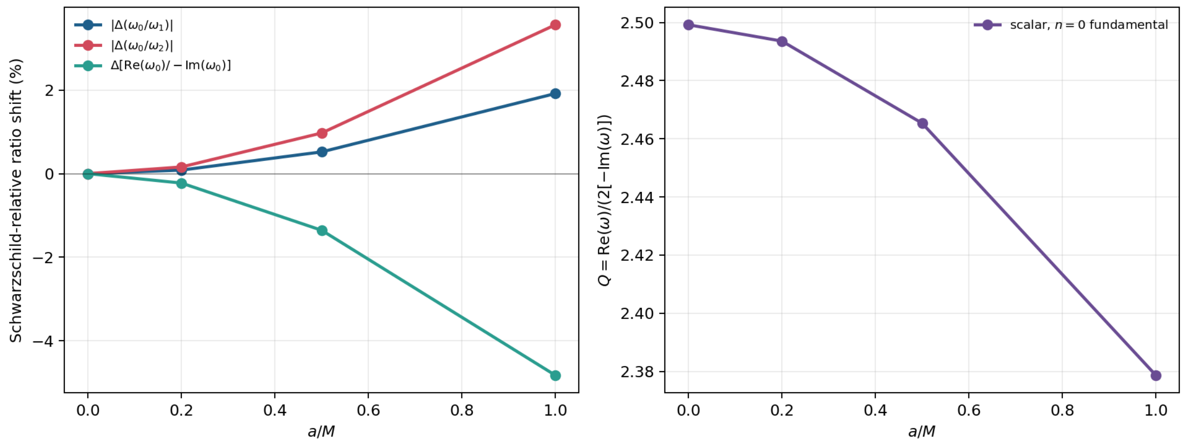

Figure 5 Scalar $\ell = 2$ dimensionless spectroscopy diagnostics. The left panel shows Schwarzschild-relative changes in ω_0/ω_1, ω_0/ω_2, and $2Q$. The right panel directly plots $Q = \mathrm{Re}\,\omega/[2(-\mathrm{Im}\,\omega)]$ for the robust fundamental branch. These quantities are catalogue diagnostics rather than an observational forecast.

The scalar $\ell = 2$ fundamental is consequently the best-controlled spectroscopic quantity in this calculation. It is a quadrupolar test-field mode, not the gravitational $\ell = m = 2$ ringdown mode measured by an interferometer. Connecting the percent-level shifts found here to data would require a gravitational waveform, detector-noise weighting, and a simultaneous treatment of mass, spin, and mode amplitudes [13, 14]. Our result is more modest but still physical: at fixed mass, the KS deformation softens the low-lying scalar spectrum throughout the sampled interval.

3.6 Comparison with existing KS literature

Our results fit naturally into the existing KS literature. Konoplya studied scalar, electromagnetic, and Dirac perturbations with WKB and time-domain methods, together with scattering and shadow observables [17]. Bolokhov, Bronnikov, and Konoplya later used a Frobenius continued fraction to show that KS overtones can respond much more strongly than the fundamental mode [18]. The present calculation confirms the comparatively smooth fundamental branches and quantifies them with a fixed-mass spectral comparison.

Direct row-by-row numerical comparison requires care because the tabulated conventions and parameter normalizations differ. In particular, much of the earlier KS QNM literature fixes the horizon radius, while the catalogue here is organized at fixed mass scale and reports a/M. For the lapse in Eq. (1), a horizon-normalized parameter $\beta = a/r_h$ corresponds to $\alpha = a/M = 2\beta/\sqrt{1-\beta^2}$, while $r_h\omega = \sqrt{4+\alpha^2}\,M\omega$. Thus the $a/M = 1$ endpoint in this paper corresponds to $a/r_h = 1/\sqrt{5}$, not to a horizon-normalized $a/r_h = 1$ table row.

A direct numerical overlap is available for the scalar $\ell = 0, n = 0$ fundamental tabulated by Konoplya at fixed $r_h = 1$ [17]. After converting $\beta = a/r_h$ to $\alpha = a/M$, our horizon-scaled frequencies agree with the published time-domain values at the 2.1×10^{-3}–2.6×10^{-3} level. Agreement with the seventh-order WKB values is weaker, 9.4×10^{-3}–1.2×10^{-2}, as expected for a low-ℓ mode [24, 11]. This comparison also makes clear why the normalization must be matched before two KS tables are compared.

$\beta = a/r_h$	$\alpha = a/M$	this work $r_h\omega$	Konoplya time-domain $r_h\omega$	Δ_{rel}
0.10	0.201008	0.221242 − 0.210676 i	0.221470 − 0.211440 i	2.603×10^{-3}
0.20	0.408248	0.222244 − 0.213397 i	0.222370 − 0.214200 i	2.631×10^{-3}
0.50	1.154701	0.229462 − 0.235795 i	0.230120 − 0.236200 i	2.343×10^{-3}

Table 6 Normalization-matched comparison with the scalar $\ell = 0, n = 0$ fixed-horizon table of Konoplya [17]. The main catalogue starts at $\ell = 2$; these rows test the parameter conversion and reproduce the published scalar fundamental once the $r_h\omega$ convention is matched.

The overtone comparison with Bolokhov–Bronnikov–Konoplya is qualitative because their published figures concern scalar $\ell = 0$ branches with $n = 5, 7, 9$, whereas our catalogue covers $\ell = 2, 3, 4$ only through $n = 2$ [18]. Both calculations nevertheless show the same hierarchy: fundamentals vary smoothly, while overtones are more deformation-sensitive and more demanding numerically. Our additional results are the fixed-mass branch comparison, the quality-factor and mode-ratio shifts, and the singular-value diagnostics.

3.7 Reported scalar rows

Table 7 collects the scalar rows used in the numerical comparison and separates the time-domain orientation from the frequency-domain values discussed in the text.

a/M	N	mode	status	$\mathrm{Re}(M\omega_{\text{fit}})$	$\mathrm{Re}(M\omega_{\text{pencil}})$	$\mathrm{Re}(M\omega_{\text{spec}})$	$\mathrm{Im}(M\omega_{\text{spec}})$
0.0	96	fundamental	high-N	0.48364248	0.48365353	0.48364387	-0.09675878
0.0	32	first overtone	cross-validated	–	–	0.46385058	-0.29560394
0.2	96	fundamental	high-N	0.48203239	0.48204305	0.48203369	-0.09665386
0.2	32	first overtone	cross-validated	–	–	0.46216559	-0.29531130
0.5	96	fundamental	high-N	0.47388656	0.47389459	0.47388693	-0.09610918
0.5	32	first overtone	cross-validated	–	–	0.45364275	-0.29379018
1.0	96	fundamental	high-N	0.44836165	0.44836213	0.44836241	-0.09424818
1.0	32	first overtone	cross-validated	–	–	0.42697149	-0.28857291

Table 7 Scalar frequency comparison. Fundamental branches are reported at $N = 96$; the cross-validated low-lying first overtones at $N = 32$ are quoted on that grid. Displayed decimals identify the branches; the supported precision is determined from spectral convergence and the continued fraction, not from the residual alone. High-N overtone tracking is retained only as an exploratory diagnostic.

3.8 Residual diagnostics

Table 8 reports scale-aware residual diagnostics associated with the same scalar rows. The raw smallest singular value answers "how close is this matrix to having a null vector?" but changes if the entire equation is multiplied by a constant. The polynomial backward error removes that arbitrary overall scale. It estimates the smallest relative perturbation of the coefficient matrices that would make the reported frequency an exact eigenvalue of the finite polynomial problem. It is

$$\eta_{\text{back}}(\omega) = \frac{\sigma_{\min}[P_N(\omega)]}{\|A_0\|_2 + |\omega|\|A_1\|_2 + |\omega|^2\|A_2\|_2}. \tag{25}$$

a/M	N	mode	dim	$\sigma_{\min}(P_N)$	η_{back}
0.0	96	fundamental	96	1.036×10^{-19}	4.542×10^{-22}
0.0	32	first overtone	32	4.271×10^{-19}	1.713×10^{-20}
0.2	96	fundamental	96	8.244×10^{-21}	3.625×10^{-23}
0.2	32	first overtone	32	3.123×10^{-19}	1.256×10^{-20}
0.5	96	fundamental	96	5.098×10^{-20}	2.272×10^{-22}
0.5	32	first overtone	32	3.598×10^{-19}	1.468×10^{-20}
1.0	96	fundamental	96	4.017×10^{-20}	1.873×10^{-22}
1.0	32	first overtone	32	1.096×10^{-19}	4.696×10^{-21}

Table 8 Residual diagnostics for the reported spectral branches. Both quantities are evaluated by SVD on the raw, unscaled polynomial matrices; the row/column equilibration used to generate eigenvalue candidates is not used here. Backward error is scale aware but is still interpreted together with spectral-size and continued-fraction convergence.

Hermiticity means $R_N = R_N^\dagger$, and positive semidefiniteness means $v^\dagger R_N v \geq 0$ for every vector v. Both follow algebraically from $R_N = P_N^\dagger P_N$, up to floating-point roundoff; they are not presented as empirical validation. The automated tests instead check their numerical implementation, Schwarzschild recovery, scaling invariance of the selected root, the inverse-Cowling axial potential, and the numerically distinct validation paths.

3.9 Local singular-value pseudospectral diagnostics of the scalar fundamental mode

An eigenvalue table tells us where a mode is located but not how sharply that location is defined. This distinction matters for QNMs because outgoing boundary conditions make the radial operator non-self-adjoint: its eigenfunctions need not be orthogonal, and small operator perturbations can move eigenvalues by much more than they would in a self-adjoint problem. A pseudospectrum addresses this by evaluating the residual throughout a region of the complex-frequency plane rather than only at the eigenvalue.

The same residual operator therefore gives a local finite-dimensional normalized singular-value pseudospectral diagnostic around a branch that has already been cross-validated by the collocation-independent continued-fraction calculation. Following the singular-value viewpoint used in black-hole pseudospectrum studies [32, 33], we define

$$\eta_N(\omega) = \frac{\sigma_{\min}(P_N(\omega))}{\sigma_{\max}(P_N(\omega))}. \tag{26}$$

The denominator removes a common matrix scale, so η_N compares the weakest and strongest stretching directions of the matrix. Small η_N means that at least one normalized wavefunction nearly satisfies the discretized equation. Contours of equal η_N are therefore level sets of "almost-solutions." A narrow set indicates a sharply localized root; a broad set indicates that a wider range of nearby frequencies produces small relative residuals. This is what the diagnostic gives us: a map of local spectral sensitivity, complementary to the single best-fit frequency. It is used here only for the finite Chebyshev matrices on the scalar $\ell = 2, n = 0$ branch.

Figure 6 shows contours of $\log_{10} \eta_N$ around the residual-minimized scalar fundamental frequencies for $a/M = 0, 0.2, 0.5, 1.0$, using $N = 64$, an 81×81 grid, and a square half-width 0.025 in the complex $M\omega$ plane. The contour centers agree with the Leaver frequencies to at worst 6.816×10^{-12} in relative difference, so the plotted basins are attached to the validated scalar branch.

The logarithm is used because the residual spans many orders of magnitude: for example, $\log_{10} \eta_N = -10$ means that the weakest matrix action is one part in 10^{10} of the strongest. The white cross in each panel marks the local residual minimum. The surrounding contour shape, not the cross, contains the new information supplied by the pseudospectrum.

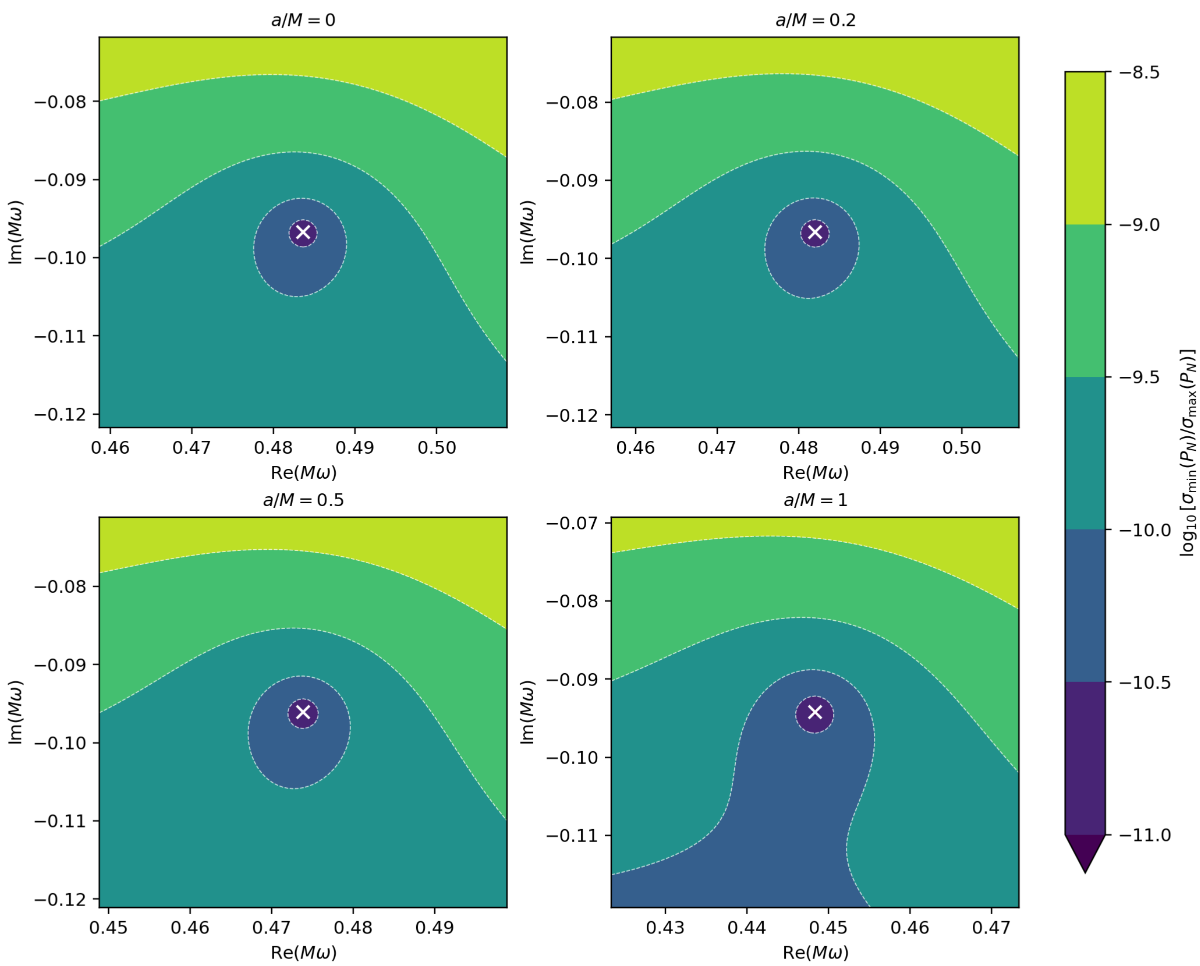

Figure 6 Finite-dimensional scalar $\ell = 2, n = 0$ singular-value pseudospectral contours for the relative residual $\eta_N = \sigma_{\min}(P_N)/\sigma_{\max}(P_N)$ at $N = 64$. White crosses mark the residual-minimized QNM locations. The panels show that the local small-η_N basin broadens as a/M increases.

Quantile and area diagnostics turn the contour picture into numbers. A p-quantile is the value below which a fraction p of all sampled grid values lies. Thus q_{10} characterizes the deepest 10% of the residual landscape, while q_{50} is its median. The area fraction below a fixed threshold measures how much of the chosen window behaves as an almost-solution. Lower-case q_p distinguishes this statistic from the QNM quality factor Q. At $N = 64$, $-q_{10}$ rises from 9.898 at $a/M = 0$ to 10.060 at $a/M = 1$, while $-q_{50}$ rises from 9.574 to 9.710. The area fraction satisfying $\log_{10} \eta_N \leq -10$ grows from 4.36% to 22.07%, a factor of 5.06. In plain terms, the low-residual neighborhood occupies more of the same local frequency window after deformation. The upper-interval contour is window-limited, so the area is used only as a local diagnostic. These diagnostics are plotted in Figure 7.

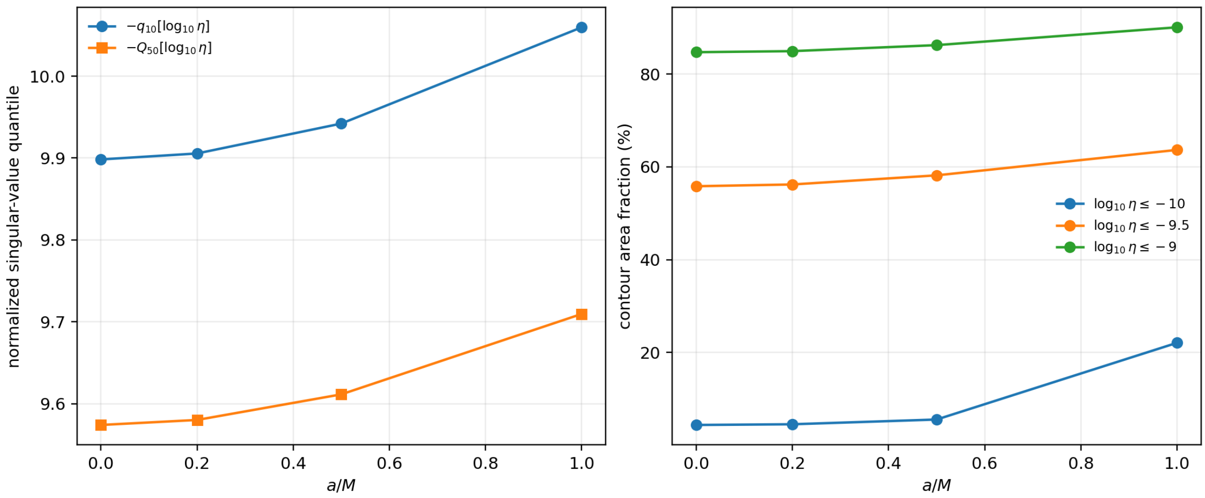

Figure 7 Scalar $\ell = 2, n = 0$ singular-value pseudospectral diagnostics at $N = 64$. Left: the normalized singular-value quantiles change with a/M. Right: fixed-threshold contour area fractions also grow; the deepest endpoint contour is window-limited and is used only as a local diagnostic.

The sign of the quantile trend is stable under the tested spectral sizes. As shown in Figure 8, the $a/M = 0$ to $a/M = 1$ gain in $-q_{10}$ is 0.097, 0.130, and 0.161 for $N = 32, 48, 64$, respectively. Thus, within the present residual normalization, scalar fundamental softening is associated with increasing local pseudospectral sensitivity. This does not mean that the mode becomes unstable: instability would require a mode with $\mathrm{Im}\,\omega > 0$. Nor is the contour width an observational error bar. It measures sensitivity of the chosen finite operator and norm.

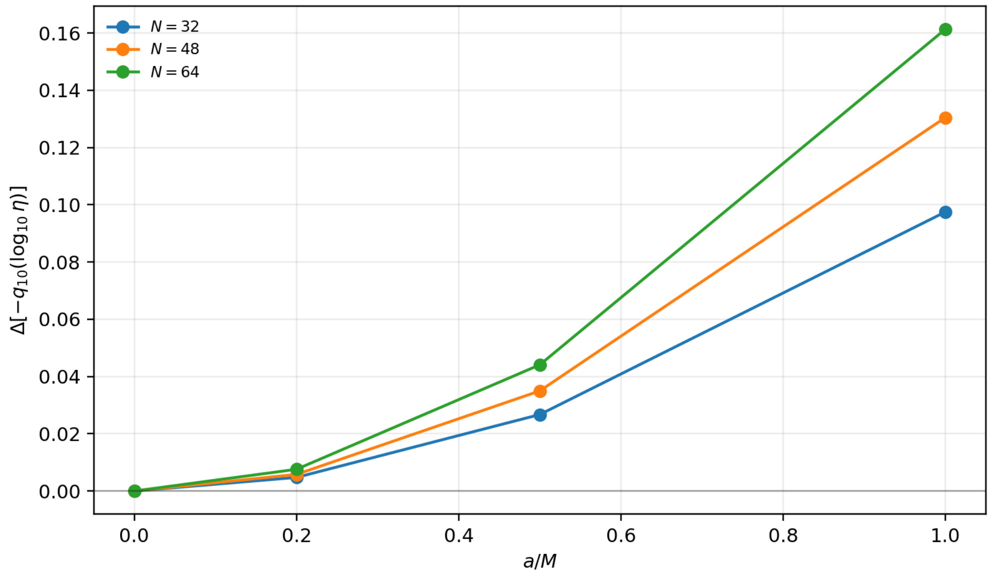

Figure 8 Finite-N check for the scalar $\ell = 2, n = 0$ singular-value pseudospectral trend. The plotted quantity is the increase in $-q_{10}(\log_{10}\eta_N)$ relative to the Schwarzschild value at the same N. The successive increase across the sampled a/M grid is stable for $N = 32, 48, 64$, although the absolute contour levels remain resolution dependent.

The dependence on grid density, frequency window, and spectral size is summarized in Table 9. It varies one numerical control at a time around the $N = 64$, 121^2-grid, half-width-0.025 reference. The endpoint change remains positive in all seven configurations and spans 0.1002–0.1647. The sign and order of magnitude are therefore robust, whereas the third decimal of a single-window result has no physical significance.

test	N	grid	half-width	$\Delta[-q_{10}(\log_{10}\eta_N)]$
grid	64	81^2	0.025	0.1613
grid	64	121^2	0.025	0.1603
grid	64	161^2	0.025	0.1607
window	64	121^2	0.020	0.1396
window	64	121^2	0.030	0.1647
spectral size	32	121^2	0.025	0.1002
spectral size	48	121^2	0.025	0.1287

Table 9 Robustness of the scalar-fundamental endpoint change in $-q_{10}(\log_{10}\eta_N)$ between $a/M = 0$ and $a/M = 1$. Every configuration gives the same positive sign and comparable magnitude. Full-precision values are generated by `scripts/audit_pseudospectrum_robustness.py`.

4 Discussion

The scalar calculation gives a direct relation between the deformed geometry and the ringing it supports. Increasing a/M at fixed mass moves both the horizon and the peak of the scalar barrier outward. At the same time the barrier becomes lower and less sharply curved in the tortoise coordinate. A wave localized near that peak therefore oscillates at a lower frequency and leaks away more slowly. This is the physical origin of the systematic softening of the scalar spectrum, rather than a numerical rescaling of an otherwise unchanged Schwarzschild mode.

For the scalar $\ell = 2$ fundamental at $a/M = 1$, the pitch decreases by 7.29% and the damping magnitude by 2.59%. Equivalently, the damping time grows by approximately 2.7%. Since the oscillation slows more than the envelope decay, the quality factor falls by 4.83%: the perturbation lives slightly longer in time but executes fewer cycles before fading. The weak ℓ-dependence follows the height and curvature of the scalar barrier as expected from WKB theory [23, 24]. These results put the qualitative softening found in earlier KS calculations on a fixed-mass, low-lying spectral footing [17, 18].

The agreement between Chebyshev and Frobenius calculations is strongest for the fundamentals and remains excellent for the first overtones at $N = 32$. The second overtones are less regular, with a maximum discrepancy of 1.202×10^{-5} on the axial $\ell = 2$ branch. This behavior is not surprising: highly damped modes are well known to require greater care in both continued-fraction and spectral calculations [19, 20, 18, 36]. For this reason the scalar fundamentals are the quantitative core of the paper, the scalar $n = 1$ modes are cross-validated low-lying first overtones at $N = 32$, and the $n = 2$ values are best regarded as indications of branch motion.

The axial calculation asks a different question. It describes how an odd-parity metric disturbance rings when the spherical quantum-corrected background is retained but its effective quantum source is not allowed to develop an independent odd-parity motion. The resulting master variable is gauge invariant, so its motion is not a coordinate artifact, but the frozen-source assumption is still physical input. The axial frequencies are therefore useful conditional predictions of that model, not universal KS gravitational-wave frequencies.

Finally, the local singular-value contours broaden with a/M on the scalar $\ell = 2, n = 0$ branch. In physical terms, nearby complex frequencies become less sharply distinguishable in the finite spectral representation as the geometry is deformed. Similar pseudospectral broadening has been used to expose the sensitivity of non-self-adjoint black-hole spectra [32, 33]. Because our calculation is finite dimensional and local in the complex-frequency plane, this is a sensitivity diagnostic, not evidence for a continuum instability.

4.1 Limitations and reproducibility

4.1.1 Interpretive boundaries

The scalar calculation requires only the KS metric and the minimally coupled Klein–Gordon equation, and is therefore the least model-dependent part of the analysis [15, 17]. The axial master variable is gauge invariant, but its homogeneous equation is obtained only after setting the invariant odd effective-source amplitudes to zero [2, 3, 4]. The axial frequencies thus belong to this inverse-Cowling model. Gauge invariance does not make the closure unique.

Nor should the percent-level shifts be read directly as detector forecasts. Part of a dimensional frequency change is degenerate with the remnant mass, and a measurement requires mode amplitudes, a waveform model, detector noise, and covariances with mass and spin [13, 10, 14]. The dimensionless ratios reported above reduce the mass degeneracy but do not replace such an analysis.

The pseudospectral plots are built from finite Chebyshev matrices rather than a controlled infinite-dimensional KS operator. Normalizing by $\sigma_{\max}$ removes an overall matrix scale, but not the dependence on discretization and norm. The robust statement is consequently limited to the observed branch: for $N = 32, 48, 64$, the endpoint quantile change has the same sign and comparable size. A continuum pseudospectrum would require a separate operator-theoretic treatment [32, 33].

Finally, the loss of monotonic Chebyshev convergence for the overtones prevents us from extending the same precision statement to arbitrary n. The reported $n = 1$ frequencies are cross-validated low-lying first overtones at $N = 32$; the $n = 2$ rows are included to show the emerging deformation trend, and higher overtones are not considered. Table 10 summarizes these distinctions.

Branch	Numerical status	Physical interpretation
Scalar $n = 0$	Robust fundamental modes	Quantitative softening and spectroscopy results.
Scalar $n = 1$	Cross-validated low-lying first overtones at $N = 32$	Secondary diagnostics; high-N tracking remains exploratory.
Overtones $n = 2$	Exploratory branch diagnostics	Qualitative deformation trends.
Axial rows	Conditional inverse-Cowling odd-parity modes	Gauge-invariant metric variable with frozen axial effective-source current.

Table 10 Numerical and physical status of the reported branches. Scalar fundamentals and first overtones are distinguished from exploratory second overtones and closure-dependent axial modes.

4.1.2 Reproducibility

The implementation is reproducible from the accompanying repository:

`https://github.com/adoolelomani2026/ks-qnm-spectral-residual-workflow`

The repository contains the Chebyshev spectral solver, the Leaver-style validation layer, catalogue-generation scripts, pseudospectrum diagnostics, generated tables, manuscript-supporting figures, and automated tests. The submission-facing commands are copied literally below.

```
pytest
python tests/test_qnm_algorithm.py --full
python scripts/run_hybrid_qnm_algorithm.py
python scripts/analyze_leaver_convergence.py
python scripts/analyze_solver_convergence.py
python scripts/analyze_scalar_potential_peaks.py
python scripts/audit_axial_model.py
python scripts/check_n128_spot.py
python scripts/analyze_pseudospectrum.py
python scripts/audit_pseudospectrum_robustness.py
python scripts/compare_konoplya2020_scalar_l0.py
```

The first two commands run the fast and complete scalar/axial validation suites. The remaining commands regenerate the main results, convergence audits, potential diagnostics, pseudospectrum data, and literature comparisons. These commands write the CSV tables, Markdown summaries, and figures under `outputs/results/` and `outputs/figures/`. The package versions used for the regenerated outputs are recorded in `outputs/results/run_metadata.json`.

5 Conclusion

The main physical result is that the KS deformation softens the low-lying test-scalar ringing at fixed mass. The deformation pushes the effective barrier outward and lowers its height and curvature. The trapped scalar wave therefore rings at a lower frequency and decays slightly more slowly. For $\ell = 2$ at $a/M = 1$, the oscillation frequency, damping magnitude, and quality factor change by -7.29%, -2.59%, and -4.83%, respectively. The scalar perturbation consequently survives for a little longer but completes fewer cycles during that time. The dimensionless ω_0/ω_1 ratio moves by 1.92%, confirming that the result is not merely a common change of mass units. This complements earlier WKB, time-domain, and Frobenius studies of the same geometry [17, 18].

At $N = 32$, the scalar fundamentals agree with the continued-fraction values to 7.92×10^{-13} or better on the sampled deformation grid. The $N = 96$ values used for the principal scalar convergence study lie on the high-resolution plateau and differ from the continued-fraction values by 1.23×10^{-11}–2.02×10^{-10}. The first overtones exhibit a finite-precision optimum near $N = 32$ and are therefore quoted as cross-validated low-lying first overtones at $N = 32$. The second overtones show larger truncation and branch sensitivity and remain exploratory. This is also why we make no claim about the highly damped spectrum, for which dedicated analytic-continuation methods are better suited [36].

For odd-parity gravitational perturbations, the gauge-invariant master variable obeys a sourced equation on a general spherical background [2, 3, 4]. Setting the invariant axial effective-source current to zero gives the inverse-Cowling potential used here, which differs from a simple lapse-deformed Regge–Wheeler potential. Its frequencies provide a useful numerical comparison and agree closely with the separate high-precision Chebyshev calculation of Batic, Dutykh, and Sukaiti [6], but their physical meaning remains conditional on the frozen-source closure. A complete nonspherical quantum theory of the KS source would be needed to remove that assumption.

The local finite-N pseudospectral basin of the scalar fundamental broadens with deformation, indicating greater local numerical sensitivity around the resonance without implying a continuum instability. The computational audits support, rather than replace, the physical conclusion: two numerically distinct radial representations identify the same robust fundamental branches, while the second overtones remain too fragile for equally strong claims [19, 28, 32].

A Schwarzschild branch-identification seeds

Table 11 records every initial target used at $a/M = 0$. Each is a literature value: no seed was selected manually and no continued-fraction result was used as an initial Schwarzschild seed. For each (s, ℓ, n), the code chooses the nearest distinct $N = 32$ Chebyshev candidate and refines it with the continued fraction. The accepted continued-fraction root then becomes the target at the next value of a/M. The machine-readable table, including a provenance field for every row, is generated as `outputs/results/schwarzschild_branch_seeds.csv`.

sector	ℓ	n	$\mathrm{Re}(M\,\omega_{\mathrm{seed}})$	$\mathrm{Im}(M\,\omega_{\mathrm{seed}})$
scalar	2	0	0.483643872211	-0.096758775978
scalar	2	1	0.463850579020	-0.295603936988
scalar	2	2	0.430544	-0.508558
scalar	3	0	0.675366	-0.096500
scalar	3	1	0.660671	-0.292285
scalar	3	2	0.633626	-0.496008
scalar	4	0	0.867416	-0.096392
scalar	4	1	0.855808	-0.290876
scalar	4	2	0.833692	-0.490325
axial	2	0	0.373671680	-0.088962320
axial	2	1	0.346710996	-0.273914876
axial	2	2	0.301053454	-0.478276983
axial	3	0	0.599443288	-0.092703048
axial	3	1	0.582643804	-0.281298113
axial	3	2	0.551684837	-0.479092791
axial	4	0	0.809178378	-0.094163961
axial	4	1	0.796631	-0.284334
axial	4	2	0.772695	-0.479900

Table 11 Literature tracking targets used to identify the Schwarzschild branches. The scalar $\ell = 2, n = 0$ value is pinned to Table I of Cavalcante and da Cunha [35]; the remaining branch targets are drawn from the Schwarzschild tabulations reviewed by Berti, Cardoso, and Starinets [10]. Rounded entries identify branches and are not treated as high-precision accuracy standards.

Acknowledgements and declarations

Funding. No funding was received to assist with the preparation of this manuscript.

Competing interests. The author declares no competing interests.

Data availability. The numerical data generated in this study, including catalogue rows, spectroscopy diagnostics, and scalar pseudospectrum grids, are preserved in release `cqg-r2-final-v3` at `https://github.com/adoolelomani2026/ks-qnm-spectral-residual-workflow/releases/tag/cqg-r2-final-v3`.

Code availability. The code used to generate the spectral results, continued-fraction validation, catalogue tables, pseudospectrum diagnostics, and figures is preserved in the same tagged release: `https://github.com/adoolelomani2026/ks-qnm-spectral-residual-workflow/releases/tag/cqg-r2-final-v3`.

Author contribution. Adel H. Al-Yoorby conceived the study, developed the numerical implementation, generated and analyzed the data, prepared the figures and tables, and wrote the manuscript.

Ethics approval. Not applicable.

Consent to participate. Not applicable.

Consent for publication. Not applicable.

Use of AI tools. OpenAI Codex, model identifier `gpt-5.6-sol`, was used during June–July 2026 to assist with code generation, debugging and refactoring, numerical-analysis scripting, algebraic cross-checks, literature discovery, manuscript drafting, language revision, and formatting. The author executed the calculations, checked cited claims against primary sources, independently reviewed the derivations and numerical outputs, and takes responsibility for the final content.

References

[1] T. Regge and J. A. Wheeler, "Stability of a Schwarzschild singularity," *Physical Review* **108**, 1063–1069 (1957). `https://doi.org/10.1103/PhysRev.108.1063`

[2] U. H. Gerlach and U. K. Sengupta, "Gauge-invariant perturbations on most general spherically symmetric space-times," *Physical Review D* **19**, 2268–2272 (1979). `https://doi.org/10.1103/PhysRevD.19.2268`

[3] C. Gundlach and J. M. Martínez-García, "Gauge-invariant and coordinate-independent perturbations of stellar collapse. I. The interior," *Physical Review D* **61**, 084024 (2000). `https://doi.org/10.1103/PhysRevD.61.084024`

[4] M. Karlovini, "Axial perturbations of general spherically symmetric spacetimes," *Classical and Quantum Gravity* **19**, 2125–2140 (2002). `https://doi.org/10.1088/0264-9381/19/8/305`

[5] P. Boonserm, T. Ngampitipan, and M. Visser, "Regge–Wheeler equation, linear stability, and greybody factors for dirty black holes," *Physical Review D* **88**, 041502(R) (2013). `https://doi.org/10.1103/PhysRevD.88.041502`

[6] D. Batic, D. Dutykh, and M. E. Sukaiti, "Quasinormal modes of the Kazakov–Solodukhin quantum-corrected black hole: a spectral analysis," *European Physical Journal C* **86**, 847 (2026). `https://doi.org/10.1140/epjc/s10052-026-16102-3`

[7] D. Batic, D. Dutykh, and M. E. Sukaiti, "KS-quantum: reproducibility repository," commit `f53435ebdb8d1124d13bee75fa54972ac1613a03`, accessed 20 July 2026. `https://github.com/dutykh/KS-quantum/tree/f53435ebdb8d1124d13bee75fa54972ac1613a03`

[8] C. V. Vishveshwara, "Scattering of gravitational radiation by a Schwarzschild black-hole," *Nature* **227**, 936–938 (1970). `https://doi.org/10.1038/227936a0`

[9] K. D. Kokkotas and B. G. Schmidt, "Quasi-normal modes of stars and black holes," *Living Reviews in Relativity* **2**, 2 (1999). `https://doi.org/10.12942/lrr-1999-2`

[10] E. Berti, V. Cardoso, and A. O. Starinets, "Quasinormal modes of black holes and black branes," *Classical and Quantum Gravity* **26**, 163001 (2009). `https://doi.org/10.1088/0264-9381/26/16/163001`

[11] R. A. Konoplya and A. Zhidenko, "Quasinormal modes of black holes: From astrophysics to string theory," *Reviews of Modern Physics* **83**, 793–836 (2011). `https://doi.org/10.1103/RevModPhys.83.793`

[12] B. P. Abbott et al. (LIGO Scientific Collaboration and Virgo Collaboration), "Observation of gravitational waves from a binary black hole merger," *Physical Review Letters* **116**, 061102 (2016). `https://doi.org/10.1103/PhysRevLett.116.061102`

[13] E. Berti, V. Cardoso, and C. M. Will, "Gravitational-wave spectroscopy of massive black holes with the space interferometer LISA," *Physical Review D* **73**, 064030 (2006). `https://doi.org/10.1103/PhysRevD.73.064030`

[14] M. Isi et al., "Testing the no-hair theorem with GW150914," *Physical Review Letters* **123**, 111102 (2019). https://doi.org/10.1103/PhysRevLett.123.111102

[15] D. I. Kazakov and S. N. Solodukhin, "On quantum deformation of the Schwarzschild solution," *Nuclear Physics B* **429**, 153–176 (1994). https://doi.org/10.1016/S0550-3213(94)80045-6

[16] T. Berry, A. Simpson, and M. Visser, "Regularity of a general class of "quantum deformed" black holes," *Universe* **7**, 165 (2021). https://doi.org/10.3390/universe7060165

[17] R. A. Konoplya, "Quantum corrected black holes: Quasinormal modes, scattering, shadows," *Physics Letters B* **804**, 135363 (2020). https://doi.org/10.1016/j.physletb.2020.135363

[18] S. V. Bolokhov, K. A. Bronnikov, and R. A. Konoplya, "Overtones' outburst and Hawking evaporation of Kazakov–Solodukhin quantum corrected black hole," *Fortschritte der Physik* **73**, 2400187 (2025). https://doi.org/10.1002/prop.202400187

[19] E. W. Leaver, "An analytic representation for the quasi-normal modes of Kerr black holes," *Proceedings of the Royal Society of London A: Mathematical and Physical Sciences* **402**, 285–298 (1985). https://doi.org/10.1098/rspa.1985.0119

[20] H.-P. Nollert, "Quasinormal modes of Schwarzschild black holes: The determination of quasinormal frequencies with very large imaginary parts," *Physical Review D* **47**, 5253–5258 (1993). https://doi.org/10.1103/PhysRevD.47.5253

[21] B. F. Schutz and C. M. Will, "Black hole normal modes: A semianalytic approach," *The Astrophysical Journal Letters* **291**, L33–L36 (1985). https://doi.org/10.1086/184453

[22] S. Iyer and C. M. Will, "Black-hole normal modes: A WKB approach. I. Foundations and application of a higher-order WKB analysis of potential-barrier scattering," *Physical Review D* **35**, 3621–3631 (1987). https://doi.org/10.1103/PhysRevD.35.3621

[23] V. Ferrari and B. Mashhoon, "New approach to the quasinormal modes of a black hole," *Physical Review D* **30**, 295–304 (1984). https://doi.org/10.1103/PhysRevD.30.295

[24] R. A. Konoplya, "Quasinormal behavior of the D-dimensional Schwarzschild black hole and higher order WKB approach," *Physical Review D* **68**, 024018 (2003). https://doi.org/10.1103/PhysRevD.68.024018

[25] H. T. Cho, A. S. Cornell, J. Doukas, and W. Naylor, "Black hole quasinormal modes using the asymptotic iteration method," *Classical and Quantum Gravity* **27**, 155004 (2010). https://doi.org/10.1088/0264-9381/27/15/155004

[26] C. Gundlach, R. H. Price, and J. Pullin, "Late-time behavior of stellar collapse and explosions. I. Linearized perturbations," *Physical Review D* **49**, 883–889 (1994). https://doi.org/10.1103/PhysRevD.49.883

[27] Y. Hua and T. K. Sarkar, “Matrix pencil method for estimating parameters of exponentially damped/undamped sinusoids in noise,” *IEEE Transactions on Acoustics, Speech, and Signal Processing* **38**, 814–824 (1990). https://doi.org/10.1109/29.56027

[28] A. Jansen, “Overdamped modes in Schwarzschild-de Sitter and a Mathematica package for the numerical computation of quasinormal modes,” *The European Physical Journal Plus* **132**, 546 (2017). https://doi.org/10.1140/epjp/i2017-11825-9

[29] D. Batic and D. Dutykh, “Quasinormal modes in noncommutative Schwarzschild black holes: A spectral analysis,” *The European Physical Journal C* **84**, 622 (2024). https://doi.org/10.1140/epjc/s10052-024-12981-6

[30] D. Batic, D. Dutykh, and B. L. Giacchini, “Unified spectral approach for quasinormal modes of Lee–Wick black holes,” *Physical Review D* **110**, 084032 (2024). https://doi.org/10.1103/PhysRevD.110.084032

[31] D. Batic and D. Dutykh, “A unified spectral approach for quasinormal modes of Morris–Thorne wormholes,” *Classical and Quantum Gravity* **41**, 215003 (2024). https://doi.org/10.1088/1361-6382/ad7cb8

[32] J. L. Jaramillo, R. Panosso Macedo, and L. Al Sheikh, “Pseudospectrum and black hole quasinormal mode instability,” *Physical Review X* **11**, 031003 (2021). https://doi.org/10.1103/PhysRevX.11.031003

[33] L.-M. Cao, J.-N. Chen, L.-B. Wu, L. Xie, and Y.-S. Zhou, “The pseudospectrum and spectrum (in)stability of quantum corrected Schwarzschild black hole,” *Science China Physics, Mechanics and Astronomy* **67**, 100412 (2024). https://doi.org/10.1007/s11433-024-2435-5

[34] M. Sadkane, “A smallest singular value method for nonlinear eigenvalue problems,” *Linear and Multilinear Algebra* **71**, 16–28 (2023). https://doi.org/10.1080/03081087.2021.2017832

[35] J. P. Cavalcante and B. Carneiro da Cunha, “Scalar and Dirac perturbations of the Reissner–Nordström black hole and Painlevé transcendents,” *Physical Review D* **104**, 124040 (2021). https://doi.org/10.1103/PhysRevD.104.124040

[36] C. Chen, J. Jing, Z. Cao, and M. Wang, “Complete quasinormal modes of type-D black holes,” *Physical Review D* **112**, 103036 (2025). https://doi.org/10.1103/f8m8-vr4l